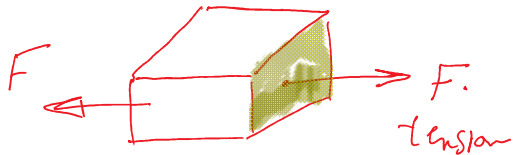
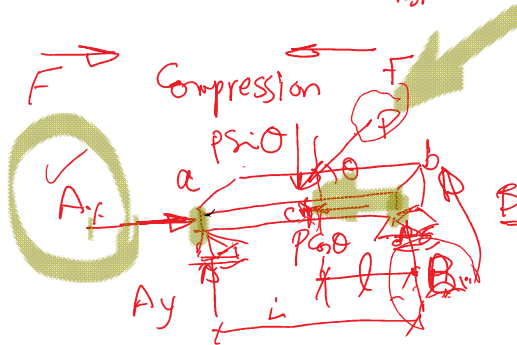
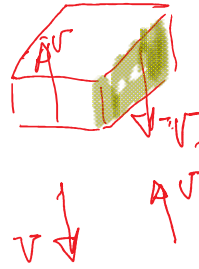


Cont. Lecture 1

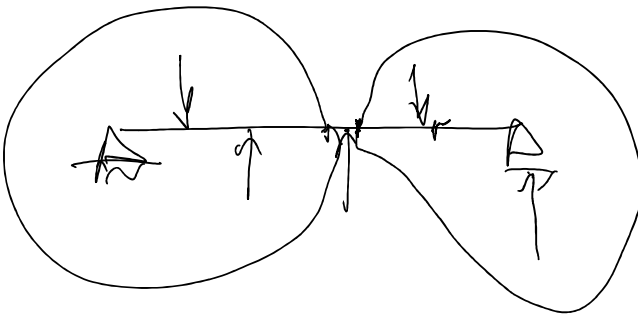
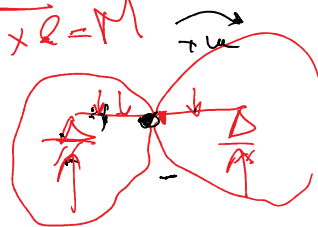
Normal force



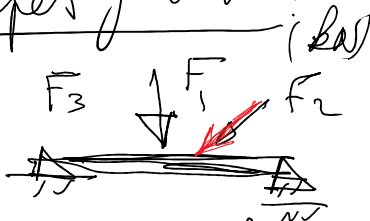
shear force



$$\frac{B.M \text{ at } C}{By \times L = M}$$

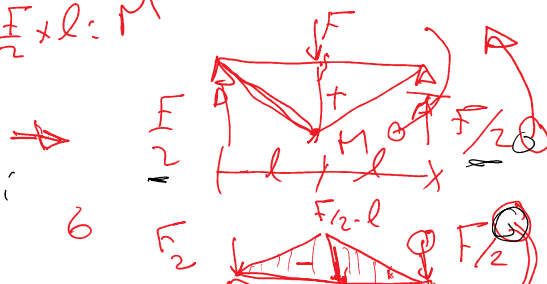


Types of loads:



Concentrated
Load(s)

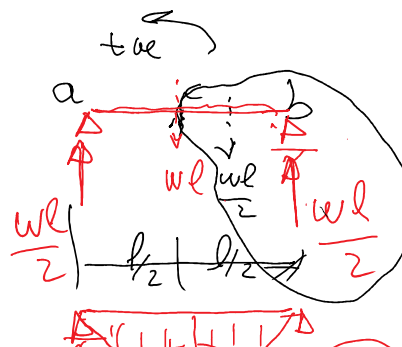
$$\frac{F}{7} \times l = M$$

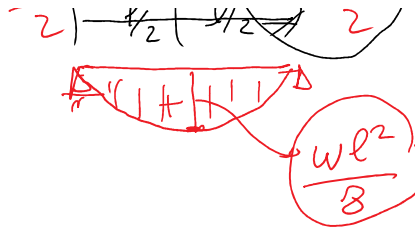
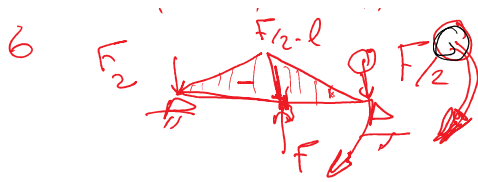


$$2 \text{ kN/m}^2$$



$$w_l = 10 \text{ kN}$$





$$B.M._C = \frac{wl}{2} \cdot \frac{l}{2} - \frac{wl}{2} \cdot \frac{l}{4}$$

$$= \frac{wl^2}{4} - \frac{wl^2}{8}$$

$$B.M._C = \frac{2wl^2 - wl^2}{8} = \frac{wl^2}{8} \rightarrow$$

Cont. Lecture 1

Stresses

Normal stress $[\sigma]$

due to

N.F. (F) M_x M_y

$$\sigma = \pm \frac{F}{A} \pm \frac{M_x}{I_x} y \pm \frac{M_y}{I_y} x$$

direct shear

shear stress

$[\tau]$

due to

Shear force V Torsion T

Transverse shear

Normal stress σ

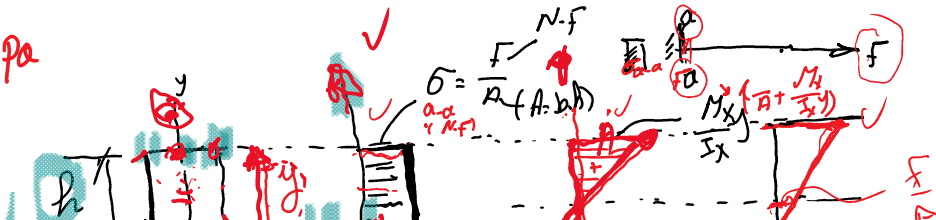
$$1 \frac{N}{mm^2} \equiv 1 \text{ MPa}$$

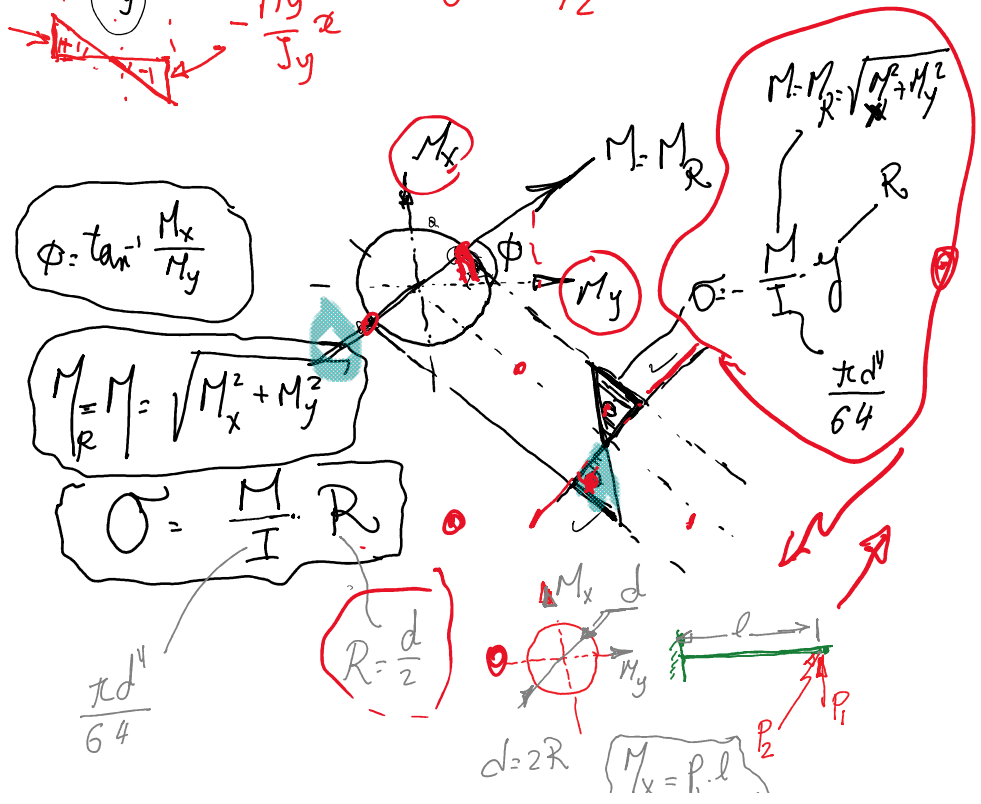
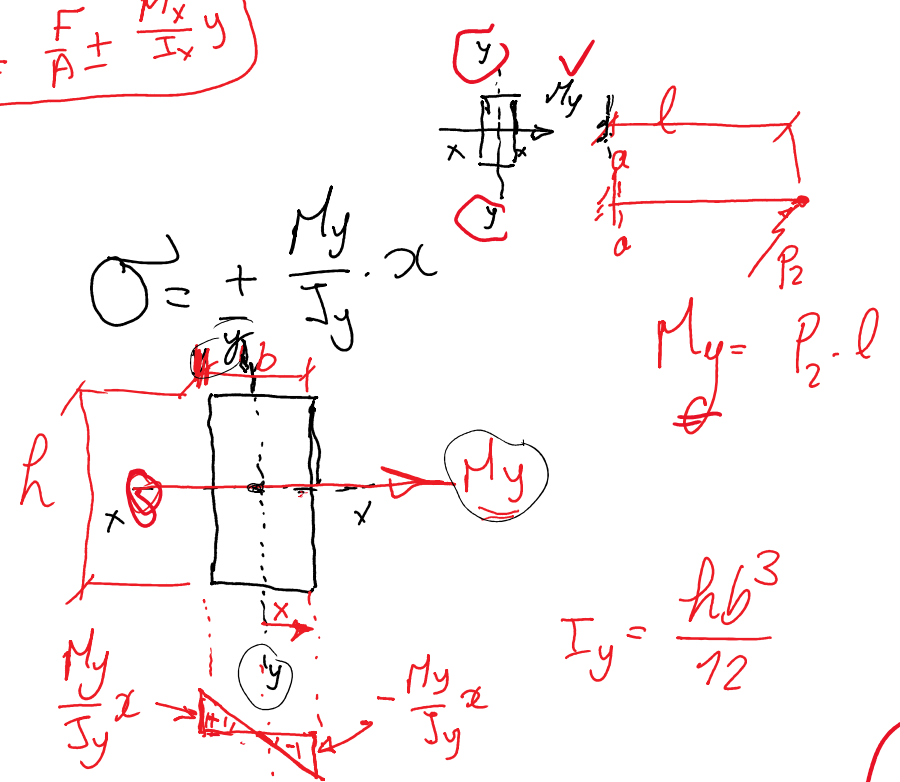
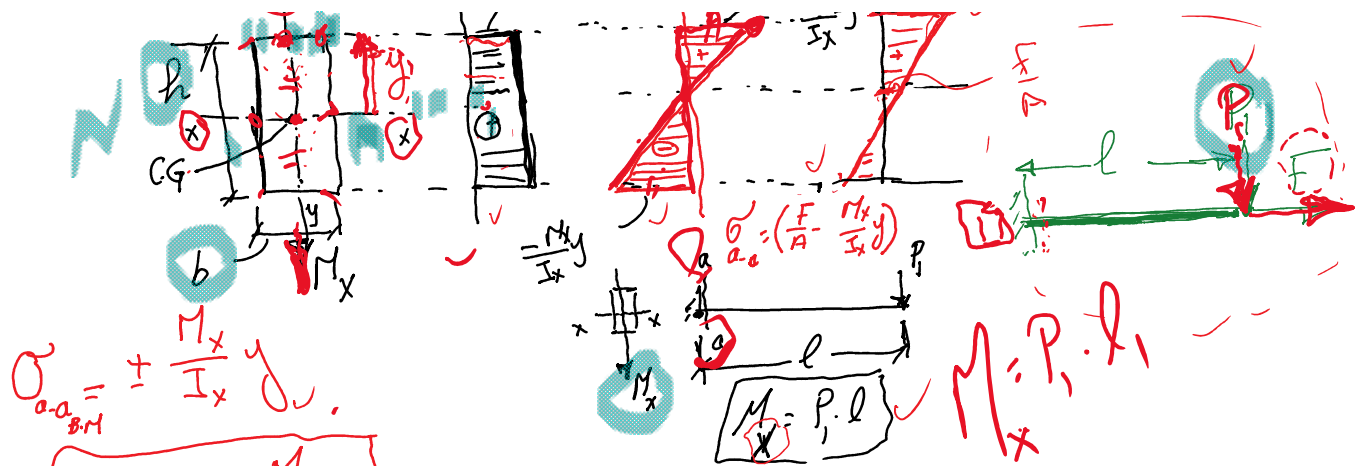
$$\sigma = \frac{F}{A}$$

Normal force $\rightarrow N$

Cross sectional area $\rightarrow mm^2$

$\frac{N}{mm^2} \hat{=} MPa$





$$\frac{1}{6^4}$$

$$d=2R$$

$$\begin{aligned} M_x &= P_1 \cdot l \\ M_y &= P_2 \cdot l \end{aligned}$$